Identical Particles

Configuration is described by

$$\psi(\vec{r}_1, \dots, \vec{r}_N) \quad (N=7 \text{ here})$$

If we exchange 1 and 2, then

$$\psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N) \longrightarrow \psi(\vec{r}_2, \vec{r}_1, \dots, \vec{r}_N)$$

$\begin{cases} \text{particle 1 in position } \vec{r}_1 \\ \text{particle 2 in } \vec{r}_2 \end{cases}$	prt. 1 in $\vec{r}_2$ prt. 2 in $\vec{r}_1$
(1)	(2)

Can these be different states?

— No! For identical particles configurations (1) and (2) represent the <u>same</u> state because particles 1 and 2 are identical (and our labeling is arbitrary!)

$$\Rightarrow \quad \psi(\vec{r}_2, \vec{r}_1, \dots \vec{r}_N) = \pm \psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N)$$

$$\begin{cases} \text{bosons} & + \\ \text{fermions} & - \end{cases}$$

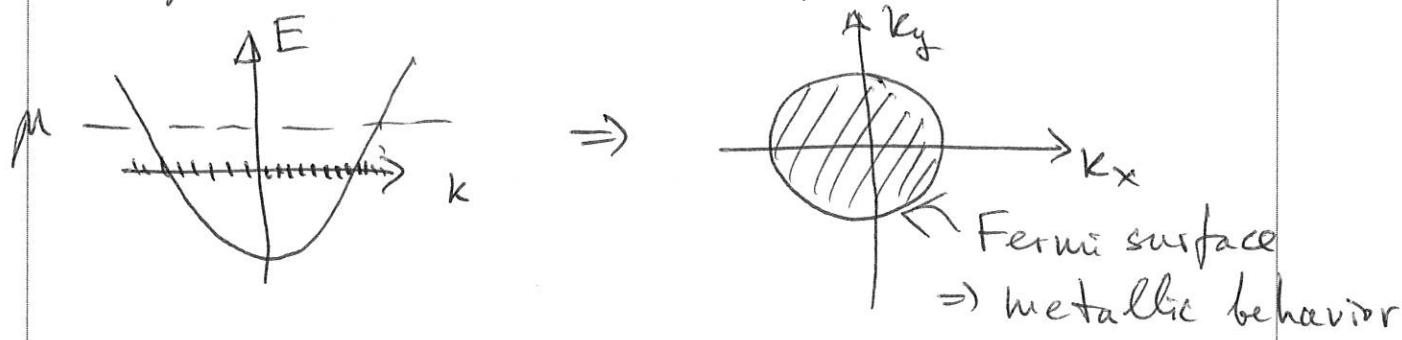
($\pm$) because exchanging 1 and 2 once again we must get back original w.f.

$$\Rightarrow \quad (\pm)^2 = +1.$$

$\Rightarrow$ <u>Pauli</u> principle for fermions is a consequence of antisymmetry of w.f.

$$\psi(\vec{r}_1, \vec{r}_1, \dots \vec{r}_N) = 0 \Rightarrow \text{one particle (per spin) in a state.}$$

This gives rise to a Fermi surface



Bosons : arbitrary number of particles can occupy the same state ⇒ Bose-Einstein condensation!

Explicit construction of symmetric/antisymmetric w.f. for $N \sim 10^{23}$ particles is very cumbersome

↳ $\boxed{\text{2nd quantization formulation offers efficient way to encode exchange statistics}}$

Examples

2 distinguishable particles : 1 in state φ_1 (φ_a)
2 in state φ_2 (φ_b)

$$\Rightarrow \psi(\bar{r}_1, \bar{r}_2) = \varphi_1(\bar{r}_1)\,\varphi_2(\bar{r}_2) = \varphi_a(\bar{r}_1)\,\varphi_b(\bar{r}_2)$$

Indistinguishable

$$\psi(\bar{r}_1, \bar{r}_2) = \frac{1}{\sqrt{2}}\left(\varphi_1(\bar{r}_1)\,\varphi_2(\bar{r}_2) \pm \varphi_2(\bar{r}_1)\,\varphi_1(\bar{r}_2)\right)$$

$$= \frac{1}{\sqrt{2}}\left(\varphi_a(\bar{r}_1)\,\varphi_b(\bar{r}_2) \pm \varphi_b(\bar{r}_1)\,\varphi_a(\bar{r}_2)\right)$$

$$\Rightarrow \psi(\bar{r}_2, \bar{r}_1) = \frac{1}{\sqrt{2}}\left(\varphi_a(\bar{r}_2)\,\varphi_b(\bar{r}_1) \pm \varphi_b(\bar{r}_2)\,\varphi_a(\bar{r}_1)\right) = \pm\psi(\bar{r}_1, \bar{r}_2)$$

Exchange operator P_{ij} : exchanges (swaps) i^{th} and j^{th} arguments in $\psi(\bar{r}_i, \bar{r}_j)$

$$P_{12} \begin{matrix} \circ & \circ & \circ \\ 1 & 2 & 3 \end{matrix} = \begin{matrix} \circ & \circ & \circ \\ 2 & 1 & 3 \end{matrix}$$

$$P_{23} \begin{matrix} \circ & \circ & \circ \\ 2 & 1 & 3 \end{matrix} = \begin{matrix} \circ & \circ & \circ \\ 3 & 1 & 2 \end{matrix}$$

$$P_{12} \begin{matrix} \circ & \circ & \circ \\ 3 & 1 & 2 \end{matrix} = \begin{matrix} \circ & \circ & \circ \\ 3 & 2 & 1 \end{matrix}$$

but $P_{13} \begin{matrix} \circ & \circ & \circ \\ 1 & 2 & 3 \end{matrix} = \begin{matrix} \circ & \circ & \circ \\ 3 & 2 & 1 \end{matrix}$

$$\Rightarrow \quad P_{12} P_{23} P_{12} = P_{13}$$

(also, $P_{23} P_{12} P_{23} = P_{13}$)

$\Rightarrow$ for fermions <u>all</u> $P_{ij} = -1$ result in consistent properties.

for bosons $P_{ij} = +1$

(Note : cannot have some $P_{ij} = -1$ and some $+1$)

Noninteracting identical particles

$$H = \sum_{i=1}^{N} \left(\frac{\vec{p}_i^{\,2}}{2m} + V(\bar{r}_i) \right)$$

- Distinguishable $\psi(\bar{r}_1 \dots \bar{r}_N) = \varphi_{\alpha_1}(\bar{r}_1)\, \varphi_{\alpha_2}(\bar{r}_2) \dots \varphi_{\alpha_N}(\bar{r}_N)$

  where $\varphi_\alpha(r)$ = α^{th} normalized eigenstate of

  $H_1 = \frac{\vec{p}^{\,2}}{2m} + V(r)$, with eigenvalue E_α.

  $$\Rightarrow E = \sum_{\alpha=1}^{N} E_\alpha .$$

- Symmetric (bosonic) w.f. of N particles

  $$\psi^S(\bar{r}_1 \dots \bar{r}_N) = \sqrt{\frac{N!}{\left(\prod_\alpha N_\alpha!\right)}} \underbrace{\frac{1}{N!} \sum_P}_{\text{symmetrization}} \varphi_{\alpha_1}(r_1)\, \varphi_{\alpha_2}(r_2) \dots \varphi_{\alpha_N}(r_N)$$

- Antisymmetric

  $$\psi^A(\bar{r}_1 \dots \bar{r}_N) = \sqrt{N!}\; \overbrace{\frac{1}{N!} \sum_P \mathrm{sgn}(P)}^{\text{antisymmetrization}} \varphi_{\alpha_1}(\bar{r}_1) \dots \varphi_{\alpha_N}(\bar{r}_N)$$

  signature of permutation P

  $$= \begin{cases} +1 & \text{for even number of exchanges} \\ -1 & \text{for odd number} \end{cases}$$

Note $P_{ij}\,\psi^{S/A}(\vec{r}_1 \dots \vec{r}_i \dots \vec{r}_j \dots \vec{r}_N) = \pm\,\psi^{S/A}(\bar{r}_1 \dots \bar{r}_j \dots \bar{r}_i \dots \bar{r}_N)$

Clearly Symmetrization (Antisymmetrization) operation makes order of indices $\{\alpha_i\}$ irrelevant $\longrightarrow$ the states are best described by <u>occupation numbers</u> N_α : $\sum N_\alpha = N$

$$E = \sum_{i=1}^{N} E_{\alpha_i} = \sum_\alpha N_\alpha E_\alpha$$

Fermion w.f. as (Slater) determinant

$$\psi^A(\bar{r}_1 \dots \bar{r}_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \varphi_{\alpha_1}(r_1) & \varphi_{\alpha_1}(r_2) & \dots & \varphi_{\alpha_1}(r_N) \\ \varphi_{\alpha_2}(r_1) & \varphi_{\alpha_2}(r_2) & \dots & \varphi_{\alpha_2}(r_N) \\ \vdots & \vdots & & \vdots \\ \varphi_{\alpha_N}(r_1) & \varphi_{\alpha_N}(r_2) & \dots & \varphi_{\alpha_N}(r_N) \end{vmatrix}$$

* so that if $\vec{r}_i = \vec{r}_j$ (different particles on the same point)
then determinant has two identical columns
$\Rightarrow \psi^A = 0$

* if $\alpha_i = \alpha_j$ (two particles occupy the same state)
then determinant has two identical rows
$\Rightarrow \psi^A = 0$ again.

<u>Example</u> free particles on a ring:

$$\varphi_n(x) = \frac{1}{\sqrt{L}} e^{ik_n x} = \frac{1}{\sqrt{L}} e^{i\frac{2\pi n}{L}x}$$

$n = 0, \pm 1, \pm 2 \ldots$

Periodic BC $\varphi_n(x+L) = \varphi_n(x)$;

$$\int_0^L |\varphi_n(x)|^2 dx = \frac{1}{L}\int_0^L dx = 1.$$

$$\hat{H}_1 = \frac{P^2}{2m} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}$$

$$\hat{H}_1 \varphi_n(x) = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\frac{1}{\sqrt{L}} e^{i\frac{2\pi n}{L}x} =$$

$$= -\frac{\hbar^2}{2m}\left(\frac{i2\pi n}{L}\right)^2 \varphi_n(x)$$

$$\Rightarrow E_n = \frac{2\pi^2\hbar^2 n^2}{mL^2}$$

1) Ground state (lowest energy state) for N bosons:

put all N into state $\varphi_0(x) = \frac{1}{\sqrt{L}}$

Occupation number $N_0 = N$

$N_{n>0} = 0.$

$E = E_0 \cdot N_0 = 0.$

$$\psi^S(x_1, x_2 \ldots x_N) = \left(\frac{1}{\sqrt{L}}\right)^N$$

2) Ground state of 2 fermions

$$\psi^A(x_1, x_2) = \frac{1}{\sqrt{2}}\left(\varphi_0(x_1)\,\varphi_1(x_2) - \varphi_0(x_2)\,\varphi_1(x_1)\right)$$

$$= \frac{1}{\sqrt{2}L}\left(e^{i\frac{2\pi}{L}x_2} - e^{i\frac{2\pi}{L}x_1}\right) =$$

$$= \frac{1}{\sqrt{2}L}\, e^{i\frac{2\pi}{L}\left(\frac{x_1+x_2}{2}\right)}\left(e^{i\frac{\pi}{L}(x_2-x_1)} - e^{i\frac{\pi}{L}(x_1-x_2)}\right)$$

$$= \frac{-2i}{\sqrt{2}L}\, e^{i K_{cm} X_{cm}} \sin\frac{\pi}{L}(x_1-x_2)$$

$K_{cm} = \frac{2\pi}{L}$ Center of mass momentum

$X_{cm} = \frac{1}{2}(x_1+x_2)$ CM position

Normalization: $\psi^A = C(-i)\frac{\sqrt{2}}{L}\, e^{i K_{cm} X_{cm}} \sin\frac{\pi}{L}(x_1-x_2)$

$$\int_0^L dx_1 \int_0^L dx_2\, |\psi^A|^2 = 1 \Rightarrow |C|^2 \frac{2}{L^2}\int dx_1 dx_2 \sin^2\frac{\pi}{L}(x_1-x_2) = 1$$

But $\sin^2\frac{\pi}{L}(x_1-x_2) = \frac{1}{2}\left(1 - \cos\frac{2\pi}{L}(x_1-x_2)\right)$

$$\Rightarrow \frac{|C|^2}{L^2}\cdot\int dx_1 dx_2 \left\{1 - \frac{1}{2}\left(e^{i\frac{2\pi}{L}(x_1-x_2)} + e^{-i\frac{2\pi}{L}(x_1-x_2)}\right)\right\} = 1$$

But $\int_0^L dx_1\, e^{i\frac{2\pi x_1}{L}} = \frac{L}{2\pi i}\, e^{iz}\Big|_{z=0}^{z=2\pi} = 0$

$\Rightarrow \frac{|C|^2}{L^2}\cdot L^2 = 1 \Rightarrow |C| = 1 \Rightarrow$ can choose $\underline{C = i}$

$N=3$: let $z_i = e^{i\frac{2\pi x_i}{L}} \Rightarrow \varphi_n(x_i) = \frac{1}{\sqrt{L}} z_i^n$

For lowest energy state we use $n=0$, $n=\pm 1$

$$E = E_0 + 2E_1$$

$$\psi^A \sim \begin{vmatrix} z_1^{-1} & z_2^{-1} & z_3^{-1} \\ 1 & 1 & 1 \\ z_1^1 & z_2^1 & z_3^1 \end{vmatrix} = \frac{z_3 - z_2}{z_1} - \frac{z_3 - z_1}{z_2} + \frac{z_2 - z_1}{z_3}$$

$$= \frac{z_2 z_3 (z_3 - z_2) + z_1 z_3 (z_1 - z_3) + z_1 z_2 (z_2 - z_1)}{z_1 z_2 z_3} =$$

$$= \frac{1}{z_1 z_2 z_3} \underbrace{(z_1 - z_2)(z_2 - z_3)(z_3 - z_1)}$$

$$(z_1 z_2 - z_1 z_3 - z_2^2 + z_2 z_3)(z_3 - z_1) =$$

$$= \cancel{z_1 z_2 z_3} - z_1^2 z_2 - z_1 z_3^2 + z_1^2 z_3 - z_2^2 z_3 + z_2^2 z_1 + z_2 z_3^2 - \cancel{z_1 z_2 z_3}$$

$$= \frac{-1}{z_1 z_2 z_3} \underbrace{\begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ z_1^2 & z_2^2 & z_3^2 \end{vmatrix}}_{(z_1 - z_2)(z_1 - z_3)(z_2 - z_3)}$$

(Vandermont determinant)

Next, $\psi^A \sim \frac{1}{z_1 z_2 z_3} z_2 \left(\frac{z_1}{z_2}-1\right) z_3 \left(\frac{z_2}{z_3}-1\right) z_1 \left(\frac{z_3}{z_1}-1\right) =$

$$= \sqrt{\frac{z_1}{z_2}}\left(\sqrt{\frac{z_1}{z_2}} - \sqrt{\frac{z_2}{z_1}}\right) \sqrt{\frac{z_2}{z_3}}\left(\sqrt{\frac{z_2}{z_3}} - \sqrt{\frac{z_3}{z_2}}\right) \sqrt{\frac{z_3}{z_1}}\left(\sqrt{\frac{z_3}{z_1}} - \sqrt{\frac{z_1}{z_3}}\right)$$

$$= \sqrt{\frac{z_1 z_2 z_3}{z_2 z_3 z_1}} (-1) \prod_{i<j} \left(\sqrt{\frac{z_i}{z_j}} - \sqrt{\frac{z_j}{z_i}}\right)$$

$$= (2i)^3 \sin\frac{\pi}{L}(x_1-x_2) \sin\frac{\pi}{L}(x_2-x_3) \sin\frac{\pi}{L}(x_3-x_1)$$

Observe: $\psi^A(x_1 x_2 x_3)$ 1) is <u>odd</u> under any pair exchange

2) is <u>zero</u> when any two coordinates coincide

3) is periodic in L: $x_j \to x_j + L$ gives the same function back.

$$\psi^A(x_1, x_2, x_3) = \frac{C}{L^{3/2}} \sin\frac{\pi}{L}(x_1-x_2) \cdot \sin\frac{\pi}{L}(x_2-x_3) \cdot \sin\frac{\pi}{L}(x_1-x_3)$$